ATM Transaction Sequence Simulation

Aim: To process deposits and withdrawals, maintain the ATM balance, enforce withdrawal limits, reject invalid transactions, and display a transaction summary.

1. Computational Thinking

1. Decomposition

- Read the initial ATM balance.
- Accept multiple deposit and withdrawal transactions.
- Validate each transaction.
- Update the balance.
- Generate the final transaction summary.

2. Pattern Recognition

- Deposits increase the balance.
- Withdrawals decrease the balance.
- Invalid transactions do not change the balance.
- Every transaction follows the same validation process.

3. Abstraction

- Consider only transaction type, amount, balance, and withdrawal limit.
- Ignore unnecessary customer details.

4. Algorithmic Thinking

- Use loops to process transactions.
- Use conditions to validate transactions.
- Use lists and tuples to store transaction records.

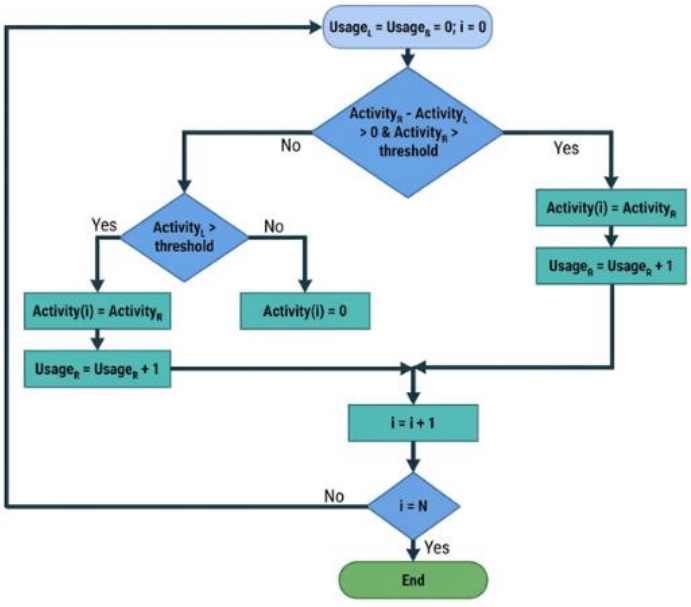
2. Algorithm

1. Start.
2. Read the initial ATM balance.
3. Set withdrawal limit and transaction counters.
4. Read the number of transactions.
5. Repeat for each transaction:

- Read transaction type and amount.
- Check whether the amount is positive.
- If deposit, add amount to balance.
- If withdrawal:
  - Check withdrawal limit.
  - Check sufficient balance.
  - If valid, subtract amount.
  - Otherwise, reject the transaction.
- Store transaction details.

6. Calculate total deposits and withdrawals.
7. Display final balance.
8. Display accepted and rejected transactions.
9. Stop.

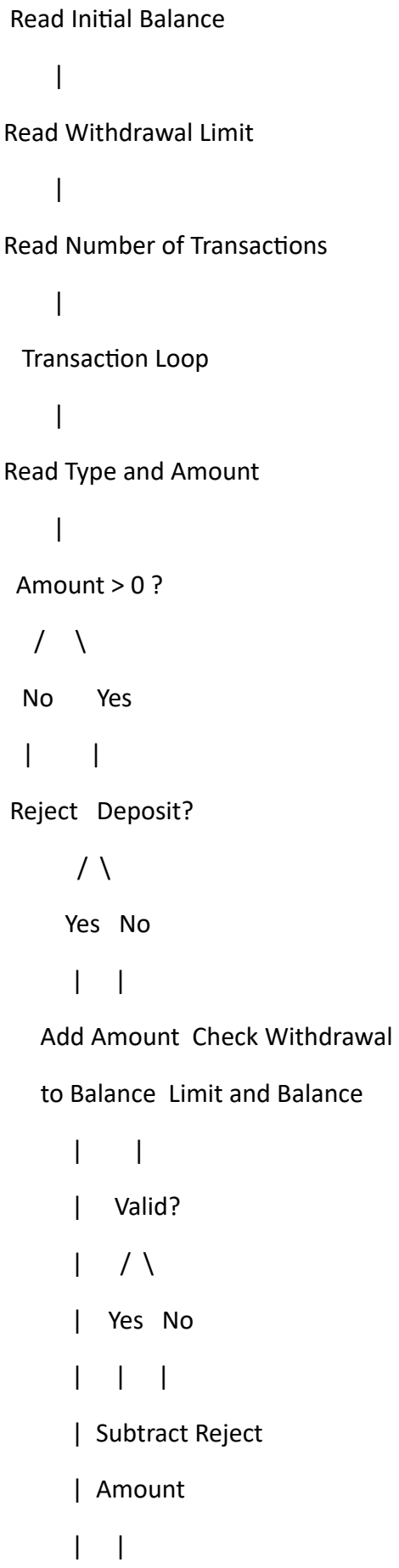
3. Flowchart:

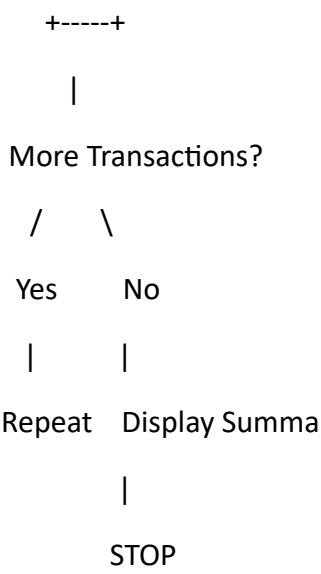


Flowchart sequence to draw:

START

|


Read Initial Balance
Read Withdrawal Limit
Read Number of Transactions
Transaction Loop
Read Type and Amount
Amount > 0 ?
No
Yes
Reject
Deposit?
Yes
No
Add Amount to Balance
Check Withdrawal Limit and Balance
Valid?
Yes
No
Subtract Reject Amount



4. Variables

Variable	Purpose
balance	Current ATM balance
amount	Transaction amount
limit	Maximum withdrawal amount
n	Number of transactions
i	Loop counter
deposits	Total deposits
withdrawals	Total withdrawals
accepted	Accepted transaction count
rejected	Rejected transaction count
transactions	Stores transaction records

5. Data Types

Data type	Variables
Integer	n, i, accepted, rejected
Float	balance, amount, limit
String	transaction_type, status

List	transactions
Tuple	Individual transaction record
Boolean	valid

6. Operators Used

Operator	Purpose	Example
+	Addition	balance + amount
-	Subtraction	balance - amount
>	Greater than	amount > 0

| <= | Less than or equal | amount 0 and amount withdrawal_limit THEN
status = "LIMIT EXCEEDED"
rejected = rejected + 1

ELSE IF amount > balance THEN
status = "INSUFFICIENT BALANCE"
rejected = rejected + 1

ELSE
balance = balance - amount
withdrawals = withdrawals + amount
accepted = accepted + 1
status = "ACCEPTED"
END IF

ELSE
status = "INVALID TYPE"
rejected = rejected + 1
END IF

STORE (transaction_type, amount, status)

END FOR

DISPLAY balance
DISPLAY deposits
DISPLAY withdrawals
DISPLAY accepted
DISPLAY rejected
DISPLAY transactions

STOP

13. Sample Input and Output

Input:

```text
Initial balance = 5000
Withdrawal limit = 3000
Number of transactions = 4

D 2000
W 1000
W 4000
W 500
```

Output:

Transaction	Amount	Status
Deposit	₹2000	Accepted
Withdrawal	₹1000	Accepted
Withdrawal	₹4000	Rejected – Limit exceeded
Withdrawal	₹500	Accepted

Final ATM Summary

Initial balance

₹5,000

Total deposits

₹2,000

Total withdrawals

₹1,500

Final balance

₹5,500

Accepted transactions
3

Rejected transactions
1

14. Conclusion

This ATM Transaction Simulation uses computational thinking, algorithms, flowcharts, variables, data types, operators, conditional statements, loops, strings, lists, and tuples to process transactions and generate an accurate transaction summary.